Hazard Analysis
Physiotherapy Companion

Team 11, technically functional
Matthew Huynh
Cieran Diebolt
Vaisnavi Shanthamoorthy
Maham Siddiqui
Eman Ashraf

Table 1: Revision History

Date	**Developer(s)**	**Change**
September 29, 2025	Maham S, Matthew H	Initial table input
October 4, 2025	Matthew H	Added initial draft of HA
October 5, 2025	Maham S	Table additions
October 8, 2025	Matthew H	Table mods
October 9, 2025	Maham S	Final edits

Contents

1 Introduction

The team's definition of a hazard is an "unsafe" condition that may lead to an unintended event that causes an undesirable outcome [Ippolito and Wallace, 1995]. In relation to software development, hazards can be understood as the underlying design decisions that result in potential harm. If these hazards remain unaddressed prior to development, they may become a source of risk. Hazards can include technical failures, privacy/security, usability and accessibility risks.

The project's goal is to develop a tool that provides guidance for a physiotherapist's prescribed home exercises. The user's form is analyzed and prompts for adjustments are given. While this can increase the rate and efficacy of recovery, it introduces several hazards. See the Section 3 in the Problem Statement document for more on project goals. The purpose of this document is to identify those hazards, outline assumptions and propose solutions to limit their presence and their frequency.

2 Scope and Purpose of Hazard Analysis

This document outlines the hazards that may occur while operating the tool. The key areas for scope include:

- **Safety-Critical Technical Hazards:** This involves failures in the underlying logic or analysis that directly impacts patient's safety. This includes incorrect identification of exercise planes, improper landmark detection, or inaccurate joint angle calculations. These failures may result in misleading real-time feedback, potentially causing the user to perform unsafe movements which may lead to re-injury.
- **Operational Technical Hazards (Application Performance):** Hazards that affect the application's functionality without risk of patient injury. These include system latency, inability to process footage due to hardware constraints, or database synchronization errors. While these impact the user experience and data integrity, they do not cause immediate harm.
- **Usability Hazards:** Usability ensures that the tool has the potential for successfully being adopted. All of the inputs to the system, such as the recording of the exercise, are generated from the user. A user-friendly interface guarantees ease of use, which encourages utilization of this tool. On the other hand, incorrect, on-screen prompts may cause confusion while the user performs the exercise. This may reduce usability and lead to ambiguous guidelines.
- **Privacy/Security Hazards:** Unnecessary data collection and storage increases the potential of unauthorized access and misuse of personal information. Such situations, like data breaches, may lead to catastrophic outcomes, which will result in lower adoption rate.

The purpose of this document is to identify risks and their associated impact to determine ways to mitigate them and identify any future occurrences

3 System Boundaries and Components

To better understand the application, we divided the application into 11 different modules. These modules are not necessarily separate entities but they are different components of the application that work together.

- Module 1: Account Creation
  - Manages patient and physiotherapist account registration, including authentication through Firebase Authentication. It ensures that only authorized users can create accounts and that account data is stored securely.
- Module 2: Valid User
  - Authenticates returning users and validates their credentials. This module handles login attempts, session management, and determines whether a user is authorized to access the application.
- Module 3: User Account
  - Manages user profile data, account settings, and session behavior including logout timing. This module ensures that user preferences are respected and that accounts remain secure.
- Module 4: Exercise Data
  - Stores and retrieves exercise programs assigned by the physiotherapist to the patient. This module handles how exercise information (repetitions, sets, range of motion targets) is organized and accessed.
- Module 5: User Activity
  - Tracks patient progress over time, including completed repetitions, Rate of Perceived Exertion (RPE), and pain levels. This module consolidates activity history for reporting purposes.
- Module 6: Analysis Control Flow
  - Controls the end-to-end pipeline from video capture through body detection to form analysis.
- Module 7: Analyze
  - Runs the computer vision model on captured video frames to detect body landmarks. This module processes the visual input and identifies key joint positions for form evaluation.

- Module 8: Feedback
  - Generates real-time audio and visual cues to guide patient form correction. This module translates analysis results into instructions for the user.
- Module 9: Metrics
  - Calculates joint angles from prescribed form. This module performs the quantitative analysis of exercise performance.
- Module 10: Draw
  - Draws the landmarks and visual feedback on the device screen during exercise. This module handles the graphical representation of body landmarks and form indicators.
- Module 11: UI
  - Manages all user-facing screens, navigation, and user input collection. This module controls the display of dashboards, exercise interfaces, and report views.

These system boundaries are included within the scope of the hazard analysis since they make up the core functionality of the application.

4 Critical Assumptions

- Assume that the user only has mobility issues pertaining to the knee area and with single joint activity.
  - Moreover, it is assumed that the user will perform an exercise that resembles the exercise.
    - This assumption is made since the application is designed for post-operative knee exercises and the computer vision library is limited to such. If a different movement is performed, the system may not be able to provide meaningful feedback.
- Assume that only one user is within the frame of the mobile device.
  - This assumption is made since the computer vision library is designed to analyze one person at a time. If multiple people are in the frame, the system may not be able to provide accurate feedback.
- Assume that the user is receptive to feedback.
  - This assumption is made since the application is designed to provide feedback to the user. If the user is not receptive to feedback then the application will not be able to provide value.

- Assume that the user is honest when providing their subjective data such as RPE and pain levels.
  - This assumption is made since the application is designed to provide feedback based on the user's subjective data. If the user is not honest then the physiotherapist may not be able to provide accurate changes to rehabilitation.
- Assume that the user has a stable connection to the internet.
  - This assumption is made since the application requires connection to the internet to access the database and receive analysis. If the user does not have internet connection then they would not be able to use the application.
- Assume that the user is undergoing physiotherapy and was advised to perform the exercise at home.
  - This assumption is made since the application is designed for patients who are undergoing physiotherapy and were advised to perform the exercise at home. If the user is not undergoing physiotherapy or was not advised to perform the exercise at home then the application should not advise them
- Assume that the device being used meets the minimum requirements for our system to function.
  - This assumption is made since if the user does not have a device that can support the application then they would not be able to use it.
- Assume that the physiotherapists using the application are licensed.
  - This assumption is made since if the physiotherapist is not licensed then they should not be advising a patient on their rehabilitation.

5 Failure Mode and Effect Analysis

Table 2: Failure Modes and Effects Analysis

Design Function	Failure Modes	Effects of Failure	Risk	Causes of Failure	Recommended Action	SR	Ref.
Authenticate user	Unauthorized access to different patient data	Breach in privacy to personal information	High	A. Having a weak password	A. Users are required to create certain password requirements to create an account and sign in.	A. FR1.4 B. SR1 C. NFR 2.3	H1-1
				B. No form of 2FA to protect personal data	B. Users are required to enter an email when creating account for 2FA		
				C. Application keeps user signed in constantly	C. The system logs the user out after 10 minutes of inactivity		

Continued on next page

Table 2 – continued from previous page

Design Function	**Failure Modes**	**Effects of Failure**	**Risk**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Authenticate user	Unable to access user profile	User unable to log into app	Medium	A. Technical failure (eg database access down) B. Too many failed login attempts	A. Pre-release verification of external library and database connectivity B. Change password option by sending email	A. SR2 B. SR3	H1-2
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	App is unable to provide feedback	High	A. Environment around user interferes with system detection B. App is unable to use the camera C. Device with insufficient charge	A. Prompt user to change to controlled environment without obstructions B. Close other apps that may be using camera C. Prompt user that sufficient charge is needed to proceed	A. SR4 B/D. SR5 C. SR6 E. FR2.5	H2-1

Continued on next page

Table 2 – continued from previous page

Design Function	Failure Modes	Effects of Failure	Risk	Causes of Failure	Recommended Action	SR	Ref.
				D. App does not have camera access permission E. Application does not have sufficient data to analyze user	D. Grant permission for app to access camera E. Application supports retry attempts if form not aligned or has obstructions		
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Application unable to provide accurate feedback	High	A. Environment around user interferes with system detection B. System places markers incorrectly	A. Refer to H2-1A B. Refer to H1-2A	A. SR4 B. SR2	H2-2
Detecting user's body (input video stream)	Unable to detect user's starting form (stationary)	Limited, incorrect or no footage for processing body parts	Medium	A. Environment around user interferes with system detection	A. Refer to H2-1A	A. SR4 B. SR6 C. FR2.5	H2-3

Continued on next page

Table 2 – continued from previous page

Design Function	**Failure Modes**	**Effects of Failure**	**Risk**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
				B. Device with insufficient charge C. Application does not have sufficient data to analyze user	B. Refer to H2-1C C. Refer to H2-1E		
Detecting user's body (input video stream)	User moves out of camera frame during exercise	App loses tracking. No feedback until user re-enters frame	Medium	A. User shifts position while exercising B. Phone/camera moves during exercise	A. Display on-screen boundary guide and play audio alert B. Prompt user to place phone on stable surface before starting	A. SR12	H2-4
Analyzing exercise form (monitoring)	Metrics calculation produces incorrect angle values	User receives wrong correction. May perform unsafe movement	High	A. CV model mistakes landmarks due to occlusion or similar clothing colors	A. Provide on-screen skeleton overlay for user to verify correct tracking before starting	A. SR4 B. SR7 C. SR13	H3-1

Continued on next page

Table 2 – continued from previous page

Design Function	Failure Modes	Effects of Failure	Risk	Causes of Failure	Recommended Action	SR	Ref.
				B. Camera angle distorts perceived joint positions	B. Provide guideline overlay for phone placement		
				C. Lighting changes mid-exercise	C. Pause and re-run environment check if confidence drops		
Analyzing exercise form (monitoring)	Metrics calculation times out or returns null	App cannot provide feedback. User continues with no guidance	Medium	A. CV processing takes too long on low-end device	A. Implement frame-skipping on low-end devices. Show "Processing..." indicator	A. SR11 B. NFR1.3	H3-2
				B. Too many landmarks missing from frame	B. Prompt user to reposition so full body is visible		

Continued on next page

Table 2 – continued from previous page

Design Function	**Failure Modes**	**Effects of Failure**	**Risk**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Analyzing exercise form (monitoring)	Metrics calculation uses wrong patient baseline	Feedback does not match prescribed ROM; potential overexertion	High	A. Session token mismatched with patient ID in database B. Physiotherapist assigned wrong exercise to patient	A. Validate patient ID against session token before each query B. Require PT to confirm exercise assignment; display exercise name to patient before starting	A. SR2	H3-3
Real-time feedback	Feedback is slow and does not reach user in time	User cannot correct form in time. Delay may cause poor performance or injury	High	A. Metrics implementation delays prompt generation (eg running average needs many data points)	A. Use sliding window with max 3-frame latency for critical form violations	A. NFR1.3 B. SR11	H4-1

Continued on next page

Table 2 – continued from previous page

Design Function	Failure Modes	Effects of Failure	Risk	Causes of Failure	Recommended Action	SR	Ref.
				B. Application does not have sufficient data to analyze user	B. Refer to H3-2B		
Analyzing exercise form (monitoring)	Unable to calculate metrics properly	System provides incorrect feedback outside physiotherapist's recommendations	High	A. System does not know limitations set by physiotherapist B.	A. Allow manual input to set daily/weekly limit on load increases B. Allow manual input to set daily/weekly limit on mobility increases	A/B. SR9	H4-2
Analyzing exercise form (monitoring)	Confusing prompts contain clinical jargon	User may not be able to perform exercise as intended	Medium	A. Prompts use technical terms (eg "extend knee 15 degrees") B.	A. Provide tutorial of movement during feedback using plain language B. Refer to H4-2	A. SR10 B. SR9 C. NFR3.1	H4-3

Continued on next page

Table 2 – continued from previous page

Design Function	Failure Modes	Effects of Failure	Risk	Causes of Failure	Recommended Action	SR	Ref.
				C.	C. Design for accessibility (audio cues + visual cues)		
Analyzing exercise form (monitoring)	Confusing prompts contain clinical jargon	User may harm themselves as a result of misunderstanding	High	A. Prompts use technical terms without explanation	A. Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-4
Analyzing exercise form (monitoring)	Confusing prompts contain clinical jargon	User becomes frustrated and stops using the app	Low	A. Prompts are confusing and use jargon	A. Refer to H4-3	A. SR10 B. SR9 C. NFR3.1	H4-5
Data Management (database)	System pulls incorrect patient data	Security breach and improper access of other user data	High	A. Database query uses wrong patient ID due to session mismatch or race condition	A. Validate patient ID against session token before each database query	A. NFR2.1	H5-1

Continued on next page

Table 2 – continued from previous page

Design Function	**Failure Modes**	**Effects of Failure**	**Risk**	**Causes of Failure**	**Recommended Action**	**SR**	**Ref.**
Data Management (database)	System pulls incorrect patient data	Feedback is given based on wrong patient's history	High	A. System incorrectly takes from wrong userID	A. Refer to H1-2A	A. SR4	H5-2
Data Management (database)	System pulls incorrect patient data	Duplicate entries created in database	Medium	A. Race condition on concurrent writes	A. Implement unique constraints on (patient ID, exercise ID, timestamp)	A. NFR2.1	H5-3
Data Management (database)	Unable to access user profile	System cannot log in user	Medium	A. Refer to H1-2	A. Refer to H1-2	A. SR2 B. SR3	H5-4

6 Safety and Security Requirements

SR1: The system shall enforce a 2FA when creating an account and logging in. The system will ensure 2FA through the email address of the user stored on the database.
Rationale: Users who do are not properly authenticated could result in privacy breaches to personal and healthcare information.
Risk: Medium
Reference: H1-1

SR2: The system shall test the connectivity to external libraries and database connectivity prior to release.
Rationale: This can result in functionality issues with the application and also obtaining data from the incorrect account.
Risk: High
Reference: H1-2, H2-2, H5-2

SR3: The system shall include an option to reset the password if too many (5) attempts are made with incorrect login.
Rationale: Inability to access the user's profile can result in not having access to the application's functionality.
Risk: Medium
Reference: H2-1, H5-2

SR4: The system shall prompt the user at the start of video streaming if there are obstructions or low-light levels and give the user a chance to change their environment prior to analysis.
Rationale: System will be unable to decipher form if the surrounding environment is not correct.
Risk: High
Reference: H2-1, H2-2, H2-3, H3-1, H3-2, H3-3, H5-2

SR5: The system shall prompt the user if the application does not have access to the camera or if there is an alternative program that is using the camera.
Rationale: This can result in an inability to detect the user's form and the application will have nothing to process.
Risk: Medium
Reference: H2-1, H3-2

SR6: The system shall monitor the mobile device's battery life and prompt the user if it falls below a certain threshold to charge the device.
Rationale: The system cannot function without proper access to power.
Risk: Low
Reference: H2-1, H2-3

SR7: The system shall provide a guideline overlay when setting up the mobile

device so they are in-frame of the view and correct orientation.
Rationale: In order to align the user to the camera and detect the user's posture throughout the exercise.
Risk: High
Reference: H3-1, H3-3

SR8: The system shall support retry attempts if the patient's form has insufficient data.
Rationale: The purpose is to provide analysis and if insufficient data is collected then it will notify the user to retry.
Risk: High
Reference: H3-1, H3-3, H4-1

SR9: The system shall allow for physiotherapists to input daily/weekly limitations on the metrics measured by the application.
Rationale: Exceeding the limits may result in improper or unsafe form and cause injury.
Risk: High
Reference: H4-2, H4-3, H4-4, H4-5

SR10: The system shall provide an option to view the motion on an internal page on the application to help guide the motion during analysis.
Rationale: Without this requirement, it may be difficult to replicate suggestions in words, having a visual cue may improve accessibility.
Risk: Low
Reference: H4-3, H4-4, H4-5

SR11: The system shall adjust frame processing rate based on device performance to maintain real-time feedback latency under 500ms. On low-end devices, frame-skipping shall be implemented to prevent processing backlog.
Rationale: Excessive latency prevents users from correcting form in real time, which may lead to repeated poor form and potential injury.
Risk: High
Reference: H3-2, H4-1

SR12: The system shall display an on-screen boundary guide during exercise setup and play audio alerts (e.g., "Please center yourself") if the user moves out of the camera frame during active exercise.
Rationale: Loss of tracking means no feedback is provided, leaving the user without guidance for correct form.
Risk: Medium
Reference: H2-4

SR13: The system shall monitor landmark confidence scores during exercise. If confidence drops below a threshold (e.g., 0.6) for more than 1 second, the system shall pause feedback and re-run the environment check.

Rationale: Changing conditions mid-exercise (e.g., lighting, occlusion) can cause inaccurate landmark detection and incorrect feedback.
Risk: Medium
Reference: H3-1

7 Hazard Analysis Traceability Matrix

The following matrix showcases the dependencies between the hazards listed and all of the requirements. This can be seen in our traceability matrix linked in the URL below, which will take you to the excel spreadsheet with the matrix:
Hazard Analysis Traceability Matrix

8 Roadmap

This section outlines which safety requirements that will be implemented within the span of the capstone timeline. It was decided that the measure of severity as well as likeliness of the hazard occurring will dictate which safety requirements are implemented. Above all, the scope of the project is also considered and the outlook on developing different requirements. As a team we have acknowledged that if everything could be implemented during the timeline of the capstone than it would result in a better final product but the limitation on time and developers were taken into account. Prioritization is based on hazard risk level (severity x likelihood), core functionality impact, and development resource constraints.

Implemented Safety Requirements

The following safety requirements have been fully implemented as of this document:

- **SR2** - Verification of external library and database connectivity within test cases.
- **SR3** - Password timeout after 5 failed login attempts.
- **SR4** - Environment check (obstructions, low-light) at start of video streaming.
- **SR5** - Camera access permission prompt and conflict detection.
- **SR7** - Guideline overlay for phone placement and user positioning.

Future Implementation

The following safety requirements are deferred beyond the capstone timeline:

- **SR1** - 2FA for account creation and login. Not integral to the minimum viable product (MVP). Target: 1-2 months post-capstone.
- **SR6** - Battery life monitoring and low-charge prompts. Low risk (inconvenience only); not critical for MVP.
- **SR8** - Retry attempts when patient form has insufficient data. This is critical because without sufficient data, the application cannot provide any analysis or feedback.
- **SR9** - Physiotherapist-configurable daily/weekly limits on metrics. After discussion with supervisor, this does not apply to post-operative knee exercises where mobility gain is the goal. Requires additional business validation.
- **SR10** - Video tutorial page for motion guidance. Usability enhancement, not core functionality for MVP.
- **SR11** - Frame-skipping and latency management for low-end devices. Real-time feedback requires processing latency under 500ms; this ensures usability across target devices.
- **SR12** - On-screen boundary guide and audio alerts when user moves out of frame. Loss of tracking directly impacts feedback quality.
- **SR13** - Mid-exercise confidence monitoring and environment re-check. Advanced feature requiring additional CV integration.

Post-Capstone Timeline

- **Short term (1-2 months after capstone):** SR1
- **Medium term (2-4 months after capstone):** SR6, SR8, SR10, SR11, SR12
- **Long term (4-6 months after capstone):** SR9, SR13

Given the importance of SR1, this will likely be addressed in the short term (1-2 months after initial development). The other requirements that are outside of the capstone timeline will be implemented in the medium term (2-4 months after development) and long term (4-6 months after development).

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Matthew - Reflection

1. I think what well in this deliverable had to have been our communication. This deliverable had a lot more separate sections that required each of us to be subject matter experts on, for example I primarily worked on the HA while others primarily did the SRS. Being able to communicate effectively allowed us to divide work while also being aware of how the other deliverable was doing and if any help was needed there.
2. I think a pain point that we experienced this deliverable was the idea of the application we were building. While we wanted to keep things general, the group found it hard to move forward without a clear idea of the project and a more granular detail. This resulted in each member having their own interpretation of the project each with slight differences that it affected consistency within documents and sections. The way that we resolved this was by identifying that there was part of the application that we had discrepancy on. From there we scheduled a meeting to determine the general flow of the program. This allowed each of us to voice opinions

while also understanding each team members point of view of the project. I would confidently say that since then we resolved those issues.

3. The risks that the team had thought of before this deliverable were very basic, it wasn't until this deliverable where we were able to add requirements to the system and then identify hazards. The ones that we thought of from the SRS are identified in the FMEA table labelled FRX.X or NFRX.X. The ones that we came about during the hazard analysis were the ones labelled SRX.X. This came about by putting ourselves in the shoes of the stakeholders and trying to understand issues that they might have. Furthermore, we had asked our advisor (Kevin), who is a physiotherapist, his perspective and how he would also view pain points. This allowed us to develop a list of hazards and thus requirements to address for the application.

4. Two other risks are technical and security risk in software products. These are important to consider as if they go unchecked then they can lead to catastrophic consequences. Technical risk for example may lead to unintended functionality and serve as a disconnect between what wanted to get implemented and what was implemented. Security risk, especially when collecting data and storing it, can result to data breaches and lead to the failure of protecting sensitive data. These both can result in a loss of trust when using different software and poor rates of adoption. Furthermore they can contribute massively towards cost of the project if a hazard would occur. These considerations must be taken into account for the health of the software and success.

Maham - Reflection

1. The process of ensuring that all team members are on the same wavelength in terms of what the application is supposed to do was very interesting. Throughout the deliverable, everyone communicated their understanding well and I was impressed with how quickly we were able to sync up. In addition, all members were able to hop on and assist anyone that needed any. This was only possible because we would conduct team meetings very often and check in on where we were with our assignment, That helped stay on track with the schedule as well.

2. I think the most important pain point was the constant juggling between priorities for each member. Everyone had assignments and midterms to complete, and that required very nuanced scheduling (to do list). I work by dividing up everything I have to do into small chunks and switch between tasks after a certain amount of time. I think most members on my team were also doing something similar to this and did not leave the bulk of their work for last minute.

3. I think the risks we thought of before working on the deliverable were quite vague and unspecific (eg. user data security). Hazards and risks

were thought of, but not solution or fixes. Once we started completing the deliverable, there were some processing hazards that we had not thought of come to light. Thinking of all the things that are possible to go wrong made it imperative to address, and we started developing fixes, whether they were similar to any others already present in the table etc.

4. As mentioned above, security risks should be considered. Patients are providing the application important and private information (e.g. videos) and they should be handled ethically. A second risk that comes to mind is the potential of the application being misused. In the context of our application, we designed it so any of the physiotherapists that are registering on the app are licensed. If not, then anyone posing as a physiotherapist can potentially use the application and create liability issues for the developers.

Vaisnavi - Reflection

1. Communication throughout the team is something that was very present throughout this deliverable. Matthew was the first one to start the HA document and kept us in the loop throughout. This helped when I had to hop on for the traceability matrix as it aided in providing insight on the concept of hazard analysis and made it seem less daunting.

2. It was hard at first to visualize how the traceability matrix would be mapped out and what it would consist of. However, after talking to the professor, the TA and my teammates, it allowed myself to envision clearly what the traceability matrix required and what I would be comparing to one another.

3. The risks the team brainstormed beforehand were very broad. We thought of hazards such as issues related to the security of our data but nothing was solidified. Only once we started to work on the document was when the remaining latter ones started to come about. As we started to have more of a grasp on our project as a whole, this was when the remaining ones began to come to light.

4. In terms of types of risks that come about in software products, one principal one previously mentioned which is a very common one is data security risk. Any access to data that is not authorized is very harmful and can cause numerous issues and leaks of personal information. Another key risk is any sort of failures that happen in regards to the reliability of the system. This could involve something like a system crash, which would ultimately impact the user's activity and their overall user safety as well.

Cieran - Reflection

1. The hazard analysis was undertaken mainly by my colleagues. However, there was a lot of communication between the two teams in terms of

keeping the files in line with one another. Specifically, communication went well regarding the assumptions which were critical for the system definitions and the safety requirement pertaining to the risks and hazards included in the hazard analysis.

2. There were misalignments in understanding of the project and in certain senses what requirements met. After team meetings and clear discussions, the ambiguity has been hammered out of both the team and the documents to put the best foot forward on the system from the get-go.

3. Risks thought of before the project for the system particularly included, but were not limited to, data privacy leaks and healthcare advice misinterpretation. Data privacy is a huge thing, users want to know what their data is being used for, where it goes, who sees it, and why all of this has to be done. Keeping data private and out of sight from those who don't need to see it is a big thing to consider within the system. The risk of that data escaping is something that has been considered the entire time. As for healthcare advice, the system isn't replacing healthcare professionals and should not try to be one nor mimic the actual healthcare that professionals give, the system is intentionally only meant to provide recommendations. Staying away from healthcare involvement is a risk for the system to avoid liabilities.

4. Considerable risks within software products include security vulnerabilities and insufficient testing/poor code quality. Security vulnerabilities, as previously discussed for this system, can expose private user data, but vulnerabilities in other platforms can be far more detrimental. Developers have to be very careful with their security measures when interacting with sensitive user data, especially when data which is leaked could drastically affect the user outside the contexts of the system. Insufficient testing and/or poor code quality can make a poor implementation into a risk for user devices and performance. If the code is poorly optimized or even broken in some places, there's a risk of causing issues with a user's device.

Eman - Reflection

1. Something that went well while writing this deliverable is the coordination between members who were working on the Hazard analysis document and the SRS requirements. As there is a strong connection between the two, good communication was required to ensure consistency throughout.

2. One pain point I experienced in this deliverable was understanding to what extent the details would be beneficial, and at what point would it be detrimental to narrow down the requirements further. I was afraid of overcommitting responsibilities without knowing our capacity and capabilities for the future. This was resolved by outlining requirements that may be stretch goals. This way, instead of over committing without enough knowledge we are still noting down potential scope for our project.

3. We had previously identified technical risks like computer vision accuracy and data security concerns, as these were obvious given the project's nature. However, while writing requirements, we discovered safety-critical risks such as patients injuring themselves from incorrect form when exercising unsupervised at home, which emerged from discussions about the fit criteria for real-time feedback and led to PR-SCR 1. We also identified usability risks for elderly or non-technical users during the accessibility requirements section, realizing that poor usability could lead to treatment non-adherence and worsen health outcomes.

4. Privacy and data breach risks are critical in healthcare applications because unauthorized access to patient health information can lead to identity theft, discrimination, legal liability, and loss of patient trust in the healthcare system. Treatment efficacy and clinical outcome risks are important because inaccurate computer vision analysis or poor feedback could lead to ineffective treatment, prolonged recovery times, or patients developing compensatory movement patterns that cause secondary injuries, ultimately undermining the therapeutic goals of the application.

References

Laura M. Ippolito and Dolores R. Wallace. *A Study on Hazard Analysis in High Integrity Software Standards and Guidelines*. 1995. URL https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nistir5589.pdf. U.S. Department of Commerce, Technology Administration, National Institute of Standards and Technology.